Czech 20-month-olds interpret gender-marked adjectives without nouns:

Emerging knowledge of the gender agreement system

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https://earlygrammarlab.github.io

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Arbitrary gender

> Every noun has a gender, and there is no sense or system in the distribution; so the gender of each must be learned separately and by heart. There is no other way. To do this one has to have a memory like a memorandum-book. In German, a young lady has no sex, while a turnip has. (…) I translate this from a conversation in one of the best of the German Sunday-school books:
>
> Gretchen: Wilhelm, where is the turnip?
> Wilhelm: She has gone to the kitchen.
> Gretchen: Where is the accomplished and beautiful English maiden?
> Wilhelm: It has gone to the opera.
>
> *Mark Twain: The Awful German Language*

- Many languages have some version of noun gender, often sex-based
  - But the semantic interpretation is irrelevant for most nouns (she-turnip, he-moon)
  - Gender knowledge is needed to construct sentences correctly
- Languages differ in the transparency and complexity of the system

How children learn noun gender?

- That is the ultimate question
- Here we approach some of the first steps
- What information can children use to learn about genders?

Gender and phonology?

- Gender often has some relations to phonological features of words
  - E. g. Italian
    - -a nouns feminine (but also -ione, -à), -o masculine (but also -ore, -ma)
    - -e is ambiguous
  - Spanish
    - -a feminine, -o masculine (many exceptions, incl. nouns ending with consonants)
- Opaque systems
  - The relations to phonology exist but are complex
    - Many exceptions, various classes, dependency on exact context or word structure
    - E. g. Czech

Determiners and gender agreement?

- Obligatory articles in some languages (German)
  - If determiners are not obligatory, gender agreement is the main cue
- In gendered languages, adjectives etc. often inflect for gender
  - Agreement with the head noun (also determiners etc.), e. g. in Czech

velk-ý$_{masc}$ náklaďák	velk-á$_{fem}$ motorka	velk-é$_{neut}$ vozidlo
big truck	big bike	big vehicle

- Subject-verb agreement in some languages
  - Some verb forms may also carry the gender feature, e. g. Czech past participles

Náklaďák zastavil-0$_{masc}$	Motorka zastavila-a$_{fem}$	Vozidlo zastavil-o$_{neut}$
The truck stopped	The bike stopped	The vehicle stopped

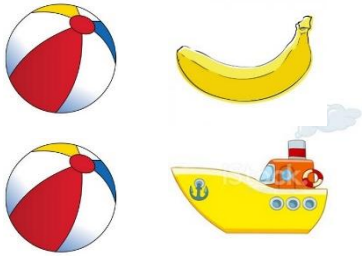
Gender in language acquisition

- Production
  - Determiners, adjectives and other elements are often delayed, hardly ever show before 2
  - Gender agreement often has nowhere to show in production
- Comprehension
  - Nouns are identified faster if preceded by an informative determiner
  - Van Heugten, Shi (2009)
    - in French, 25-mo. olds
  - Johnson (2005)
    - in Dutch 28-mo. olds
  - Lew-Williams, Fernald (2007)
    - in Spanish, 3-yr.-olds

Regarde, le ballon/la banane!

Control condition
... le ballon/le bateau

Agramatical condition
... *la ballon/*le banana

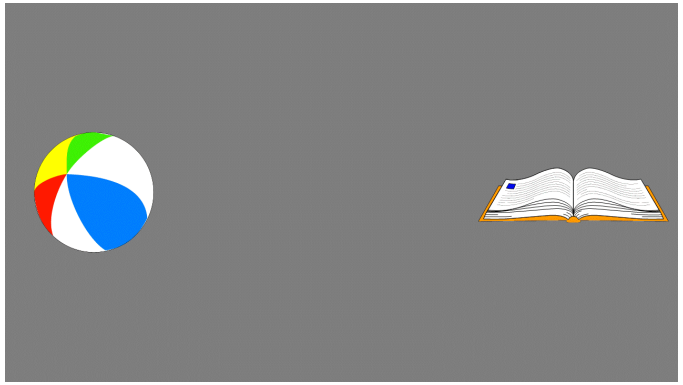


Anticipation from bound morphemes

- E. g. in Czech (Smolík, Bláhová, 2018)
  - Matching condition

    Taková$_{fem}$ veliká$_{fem}$ kniha$_{fem}$ / Takový$_{masc}$ veliký$_{masc}$ míč$_{masc}$

    such$_{fem}$ big$_{fem}$ book$_{fem}$ / such$_{masc}$ big$_{masc}$ ball$_{masc}$
  - Mismatching condition
    - Takový$_{masc}$ veliký$_{masc}$ kniha$_{fem}$ / Taková$_{fem}$ veliká$_{fem}$ míč$_{masc}$



- 21- to 24-month-olds looked at the picture based on adjective gender
  - Only those with better offline sentence scores

Gender category or morpheme association?

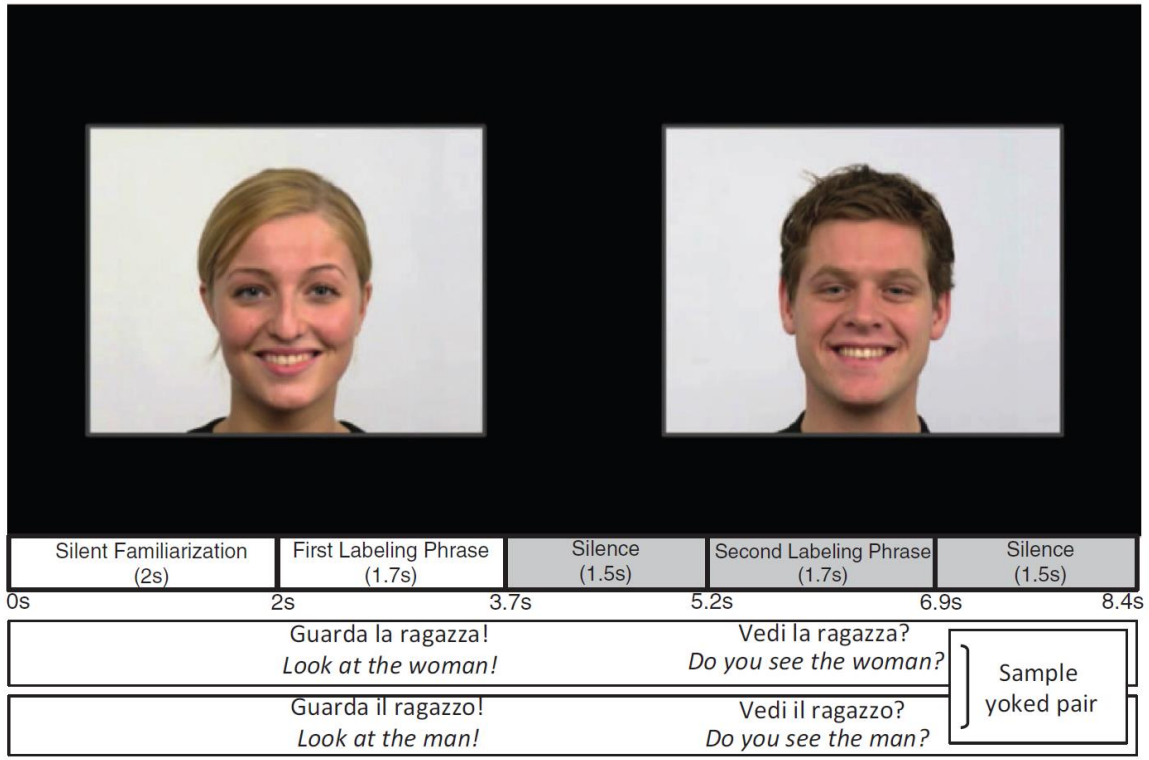
- If a French child, after hearing *le* (masc)
  - looks at *ball* ($\text{ballon}_{\text{masc}}$) and not *banane* ($\text{banane}_{\text{fem}}$) ?
- Does it mean they have the gender category?
- Could be they simply learned to associate the article *le* with the noun
  - Lexically specific association
- How can we tell?
  - What about testing the semantic interpretation?
    - Although grammatical gender is arbitrary for many words, it has semantic interpretation
  - Do children have this knowledge?

Gender interpretation in infants

- Ferry et al. (2020)
  - Tested Italian 12, 18, 24-month-olds
  - Used noun pairs such as

    ragazzo vs. ragazza

    nonno vs. nonna
  - Even 12-month-olds looked at the correct picture
- Knowledge of gender?
  - Or knowledge of specific nouns?



The questions

- Can we show interpretation of gendered forms in Czech toddlers?
  - Can we do it without noun labels?
- Do Czech children use gender information from adjectival inflections?
- Do they associate it with natural gender?
  - Adjectives only obtain gender interpretation from inflections, not meaning
  - Semantic implications of adjectival gender cannot be due to a learned meaning of a nominal label
    - Unlike in the Italian experiment
  - I. e., interpreting would suggest it is not just noun-affix association

Gender system of Czech nouns

- Nouns inflect for 7 cases and 2 numbers
  - Potentially 14 combinations, but many forms are syncretic
  - Different noun classes have between 4 and 10 distinct forms
- Three or four genders
  - Masculine (animate, inanimate), feminine, neuter
    - each has a number of distinct inflectional classes: masculine 6, feminine and neuter 4
    - these capture many but not all nouns
  - Gender can be seen as grouping of inflectional classes
    - Classes are embedded within genders

Some common noun classes

pán	muž	stroj	žena	růže	město	moře
masc. anim.	masc. anim.	masc. inan.	fem.	fem.	neut.	neut.

Some common noun classes

		pán	**muž**	**stroj**	**žena**	**růže**	**město**	**moře**
		masc. anim.	**masc. anim.**	**masc. inan.**	**fem.**	**fem.**	**neut.**	**neut.**
Nom.	sg.	pán-Ø	muž-Ø	stroj-Ø	žen-a	růž-e	měst-o	moř-e
Gen.	sg.	pán-a	muž-e	stroj-e	žen-y	růž-e	měst-a	moř-e
Dat.	sg.	pán-u / pán-ovi	muž-i	stroj-i	žen-ě	růž-i	měst-u	moř-i
Acc.	sg.	pán-a	muž-e	stroj-Ø	žen-u	růž-i	měst-o	moř-e
Voc.	sg.	pan-e	muž-i	stroj-i	žen-o	růž-e	měst-o	moř-e
Loc.	sg.	pán-u / pán-ovi	muž-i	stroj-i	žen-ě	růž-i	měst-ě / měst-u	moř-i
Instr.	sg.	pán-em	muž-em	stroj-em	žen-ou	růž-í	měst-em	moř-em
Nom.	pl.	pán-i / pán-ové	muž-i	stroj-e	žen-y	růž-e	měst-a	moř-e
Gen.	pl.	pán-ů	muž-ů	stroj-ů	žen-Ø	růž-í	měst-Ø	moř-í
Dat.	pl.	pán-ům	muž-ům	stroj-ům	žen-ám	růž-ím	měst-ům	moř-ím
Acc.	pl.	pán-y	muž-e	stroj-e	žen-y	růž-e	měst-a	moř-e
Voc.	pl.	pán-i / pán-ové	muž-i	stroj-e	žen-y	růž-e	měst-a	moř-e
Loc.	pl.	pán-ech	muž-ích	stroj-ích	žen-ách	růž-ích	měst-ech	moř-ích
Instr.	pl.	pán-y	muž-i	stroj-i	žen-ami	růž-emi	měst-y	moř-i

Adjectival system in Czech

- Adjectives agree with the word noun in gender, number, case

velk-ý$_{masc}$ náklaďák	velk-á$_{fem}$ motorka	velk-é$_{neut}$ vozidlo
big truck	big bike	big vehicle

  - Examples of cases: velk-ou motorku, velk-ého náklaďáku
- Most common type of adjectives differentiates gender

- Another adjective class has does not distinguish genders

posledn-í$_{masc}$ náklaďák	posledn-í$_{fem}$ motorka	posledn-í$_{neut}$ vozidlo
last truck	last bike	last vehicle

The experiment

- Screen with a male and female face
  - 4 yoked pairs of different faces (from RADIATE, Conley et al., 2018)

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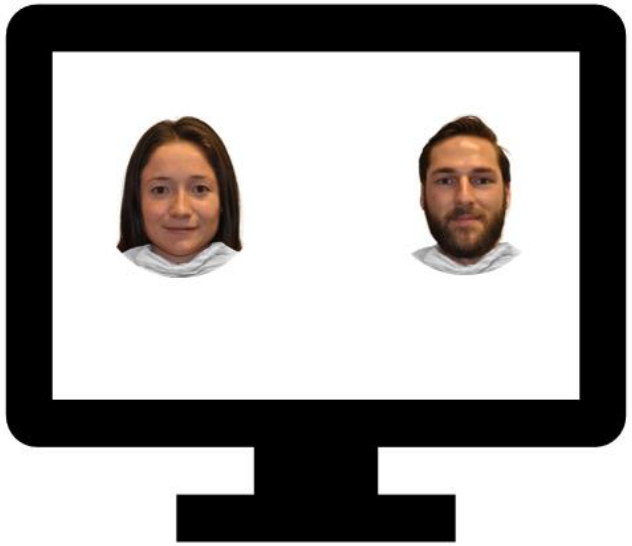
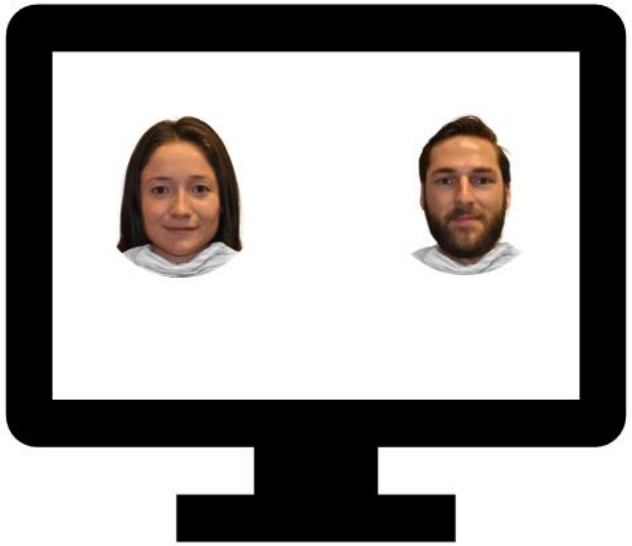


The experiment

- Screen with a male and female face
  - 4 yoked pairs of different faces (from RADIATE, Conley et al., 2018)
- Passage with two instances of an adjective
  - Jéé, hele! Koukej, jak je chytrý/chytrá! Je fakt chytrý/chytrá!
  - Yay, see! Look how smart$_{masc/fem}$ (s/he) is? (S/he) is really smart$_{masc/fem}$!



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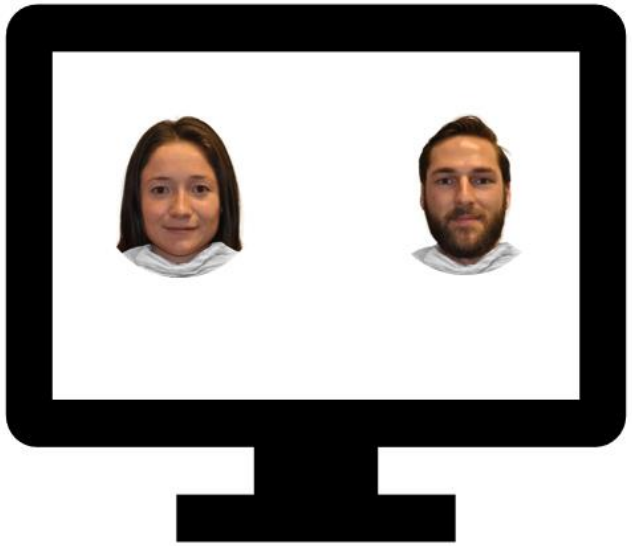
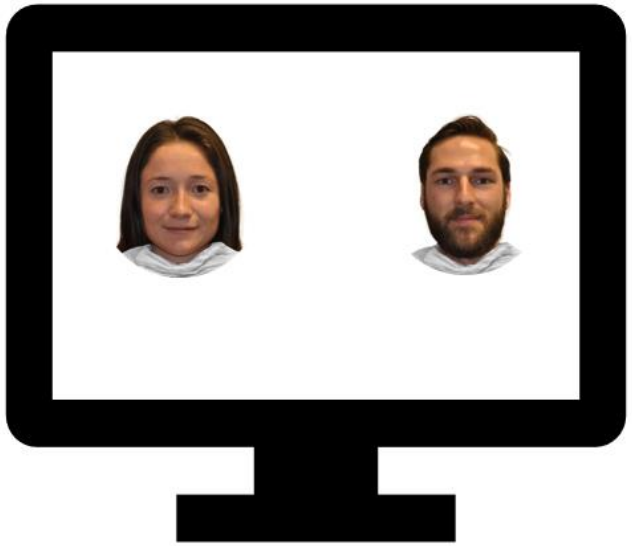
  - Yay, see! Look how smart$_{masc/fem}$ (s/he) is? (S/he) is really smart$_{masc/fem}$!
- 4 different adjectives
  - chytrý/á (clever), veselý/á (jolly, happy), smutný/á (sad), hezký/á (pretty)
- 16 trials, 8 per gender condition
  - each adjective 2 x per condition, counterbalancing female/male face location



The baseline

- Can children reliably distinguish between men and women?
- Two initial trials labelling *paní* (lady, woman) and *pán* (gentleman)
  - Jů, koukej! Tady je nějaká/ý paní/pán! Vidíš tu/toho paní/pána?
  - Hey, look! Here is some lady/man! Do you see the lady/man?



Participants and procedure

- Using Eyelink 1000+ (SR Research) with a 25mm lens
  - Target sticker on the child's forehead, faces approx. 55 cm from the lens
- Created and presented in SR Research Experiment Builder 2.44.7
  - Pictures presented on a 22-inch screen
  - Children sitting on parent's lap
  - Video animation with music played after the baseline trials and then after each set of 4 trials
- A total of 53 children aged 18 to 24 months visited the lab
  - The data were collected from 46
  - Different analyses had between 31 and 40 children, depending on cleaning criteria

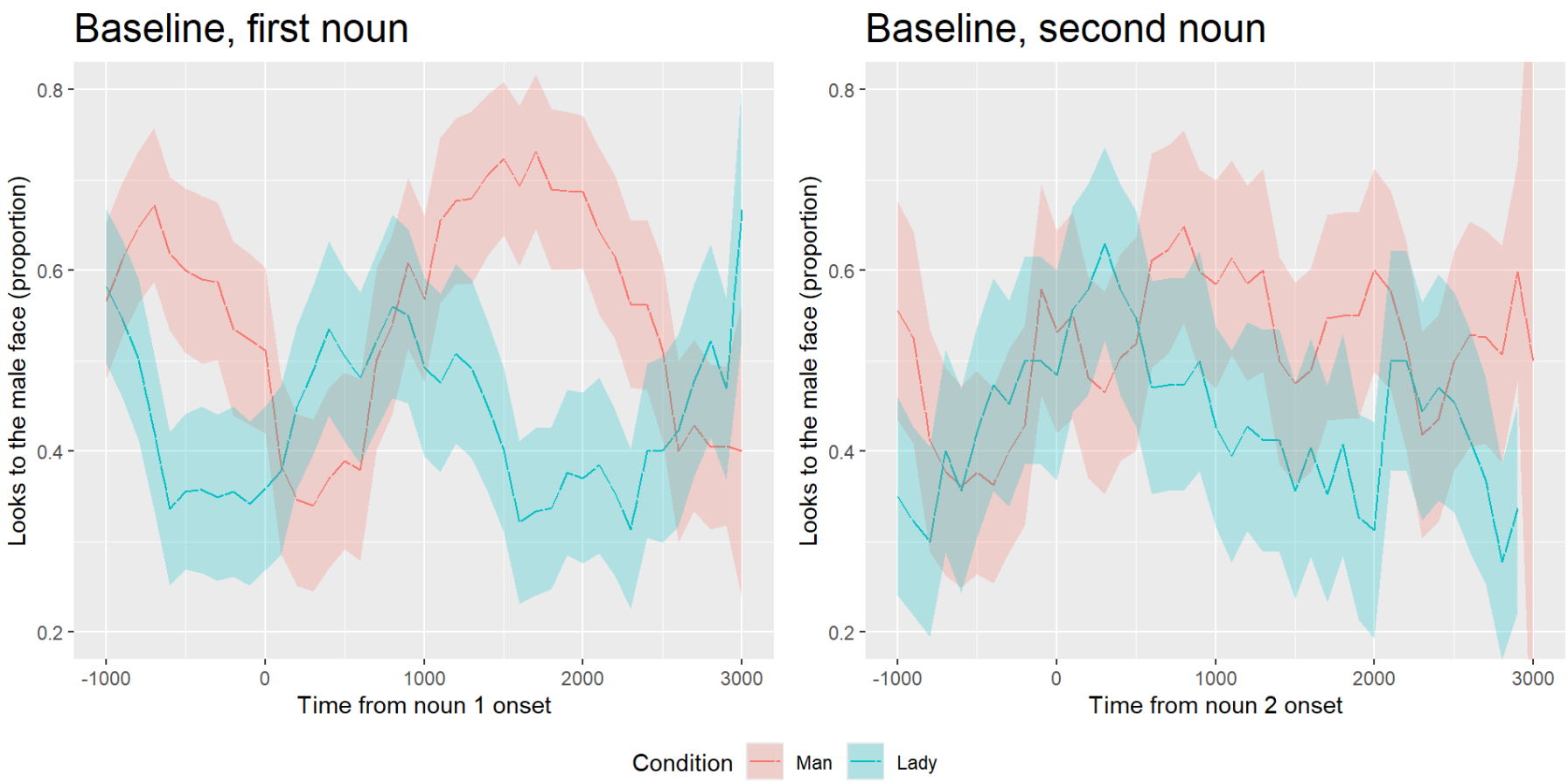
Analyses

- Two types of analyses, each separately for the first and second adjective
- Time windows
  - Pre-adjective: 1500 ms, from 1000 ms before to 500 ms after the adjective onset
  - Post-adjective: 500 to 3000 ms after the adjective onset
  - Mixed model: effects of time window and condition on the proportion of looks (straight or logit)

```
Prop ~ expcond*window + (window | pic/trial) + (window*condition | child)
```

- Permutation analysis
  - Running a glmm on each 4-ms time slice with participant and trial random intercepts
  - Examining sequences (clusters) of significant results

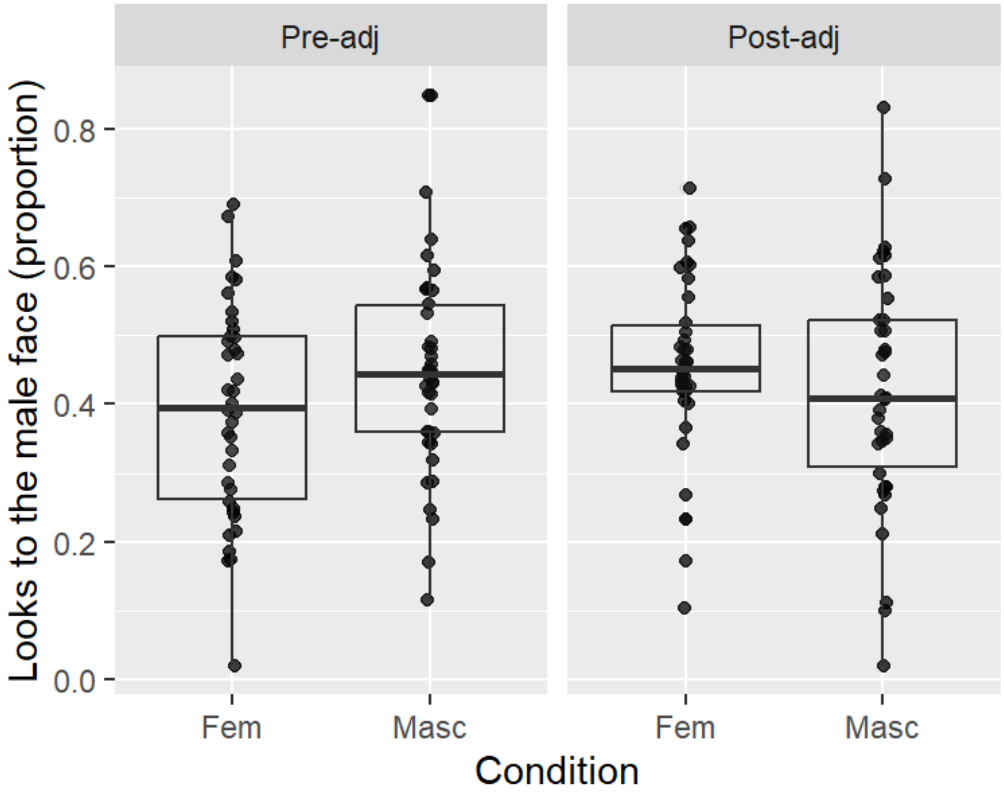
Results: baseline



- Clear differentiation on the first noun; second noun?

Results for Adjective 1

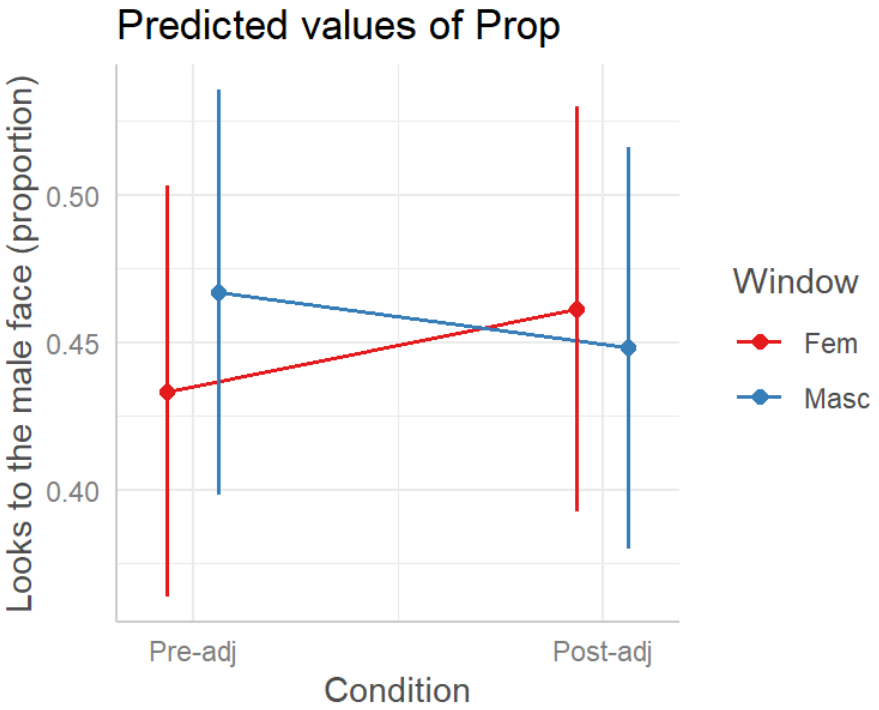
- Time windows



Results for Adjective 1

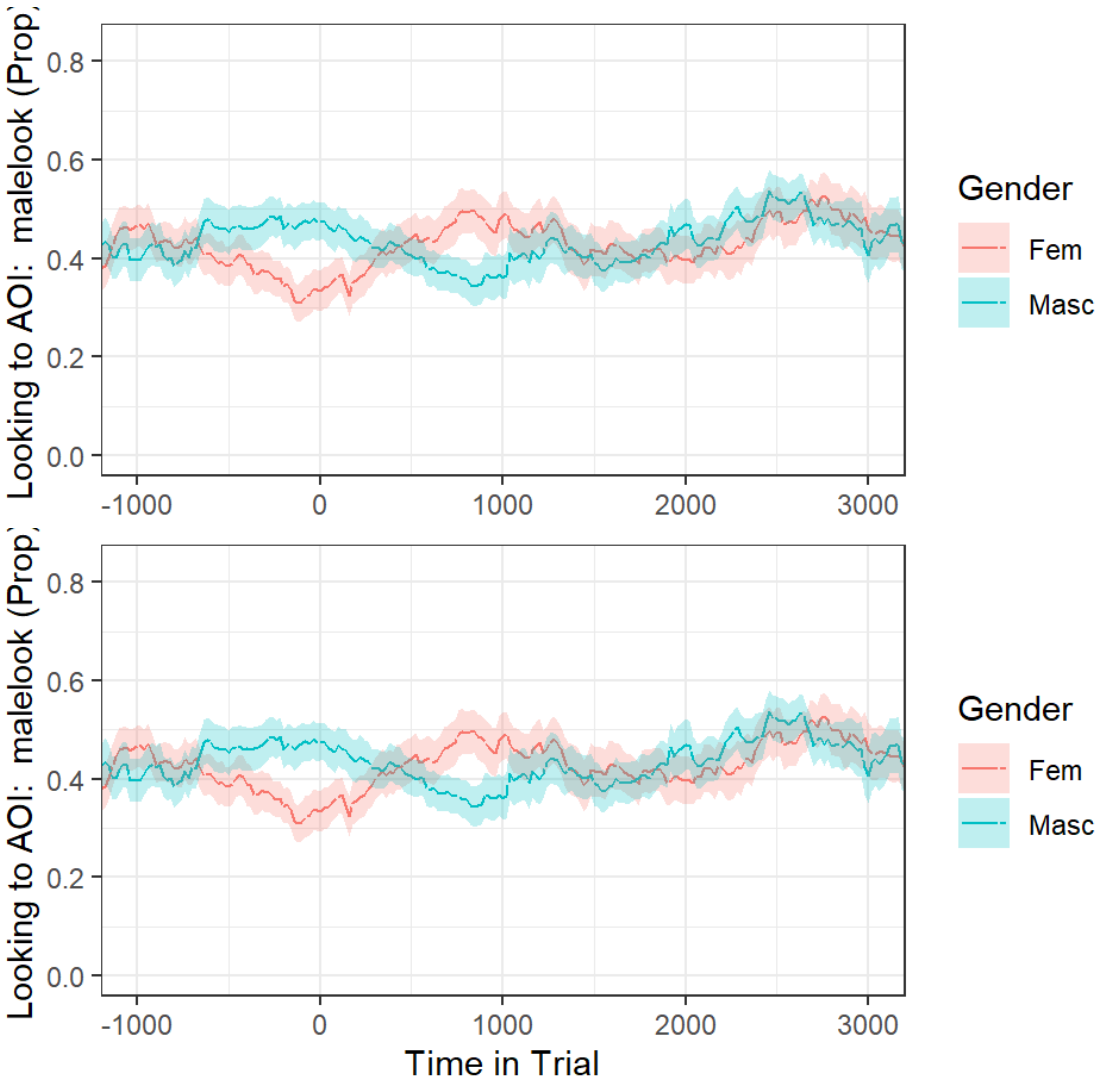
- Time windows
  - No significant interaction

	Prop			LogitAdjusted		
Predictors	*Estimates*	*Statistic*	*p*	*Estimates*	*Statistic*	*p*
(Intercept)	0.43	12.19	**<0.001**	-0.67	-1.62	0.106
Condition	0.03	1.03	0.305	0.20	0.56	0.576
Window	0.03	0.86	0.393	0.32	0.90	0.370
Condition x Window	-0.05	-1.03	0.305	-0.32	-0.64	0.523
Random Effects						
σ^2	0.12			14.02		
τ_{00}	0.00 $_{\text{Session_Name_}}$			0.07 $_{\text{Session_Name_}}$		
	0.00 $_{\text{identifier:pic}}$			0.00 $_{\text{identifier:pic}}$		
	0.01 $_{\text{pic}}$			0.81 $_{\text{pic}}$		
N	16 $_{\text{identifier}}$			16 $_{\text{identifier}}$		
	8 $_{\text{pic}}$			8 $_{\text{pic}}$		
	38 $_{\text{Session_Name_}}$			38 $_{\text{Session_Name_}}$		
Observations	903			903		
Marginal R^2 / Conditional R^2	0.001 / NA			0.001 / NA		



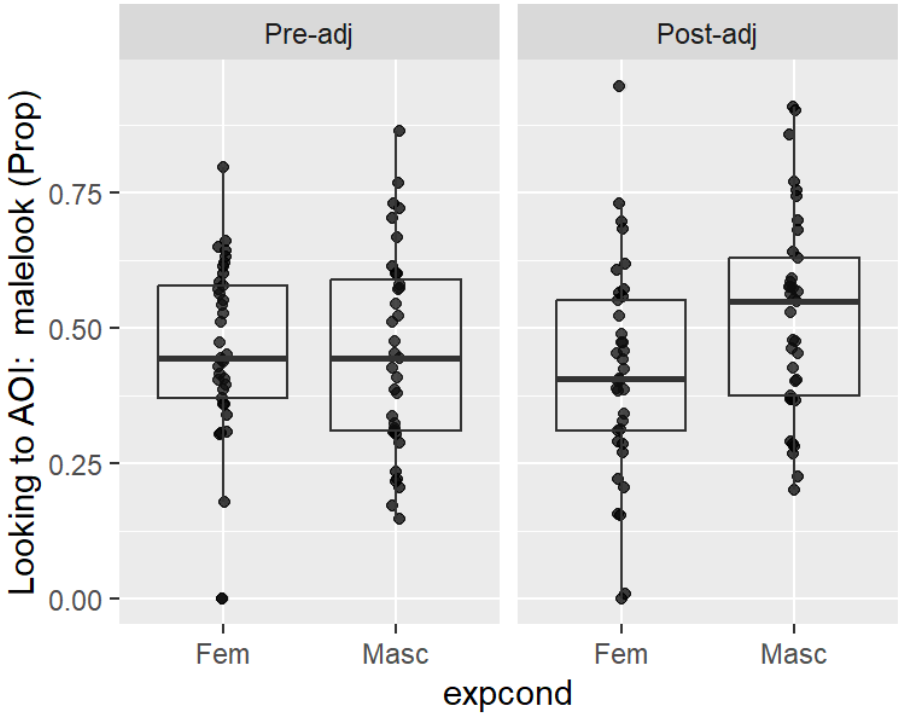
Results for Adjective 1

- Time windows
  - No significant interaction
- No significant clusters in the permutation analysis
  - Trying children with min. 2 or 3 trials/condition



Results for Adjective 2

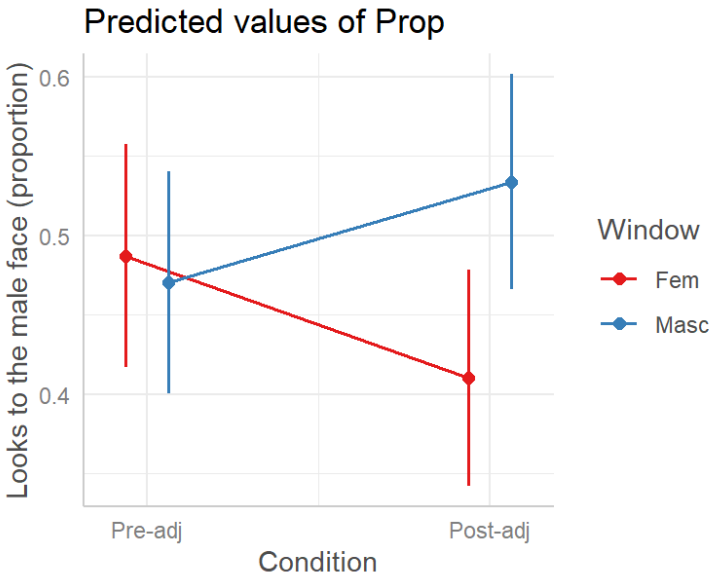
- Time windows



Results for Adjective 2

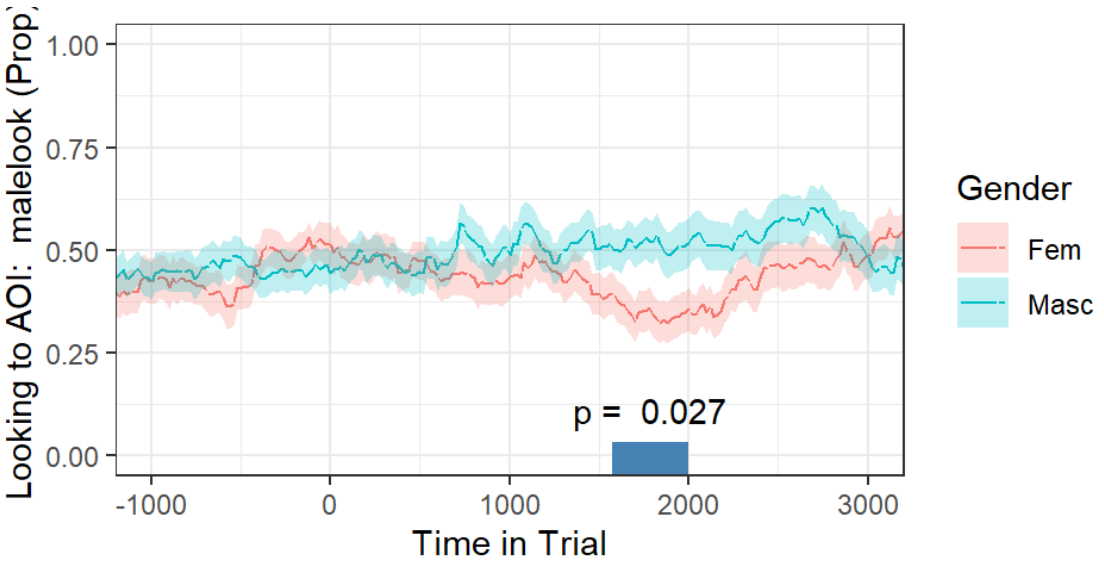
- Time windows
  - Significant interaction

	Prop			LogitAdjusted		
Predictors	*Estimates*	*Statistic*	*p*	*Estimates*	*Statistic*	*p*
(Intercept)	0.49	13.61	**<0.001**	-0.11	-0.23	0.815
Condition	-0.02	-0.41	0.680	-0.36	-0.72	0.472
Window	-0.08	-2.11	**0.035**	-1.07	-2.44	**0.015**
Condition x Window	0.14	2.74	**0.006**	1.74	2.82	**0.005**
Random Effects						
σ^2	0.13			18.13		
τ_{00}	$0.00_{\text{Session_Name_}}$			$0.28_{\text{Session_Name_}}$		
	$0.00_{\text{identifier:pic}}$			$0.19_{\text{identifier:pic}}$		
	0.00_{pic}			0.56_{pic}		
ICC	0.05			0.05		
N	$16_{\text{identifier}}$			$16_{\text{identifier}}$		
	8_{pic}			8_{pic}		
	$37_{\text{Session_Name_}}$			$37_{\text{Session_Name_}}$		
Observations	763			763		
Marginal R^2 / Conditional R^2	0.015 / 0.065			0.014 / 0.067		



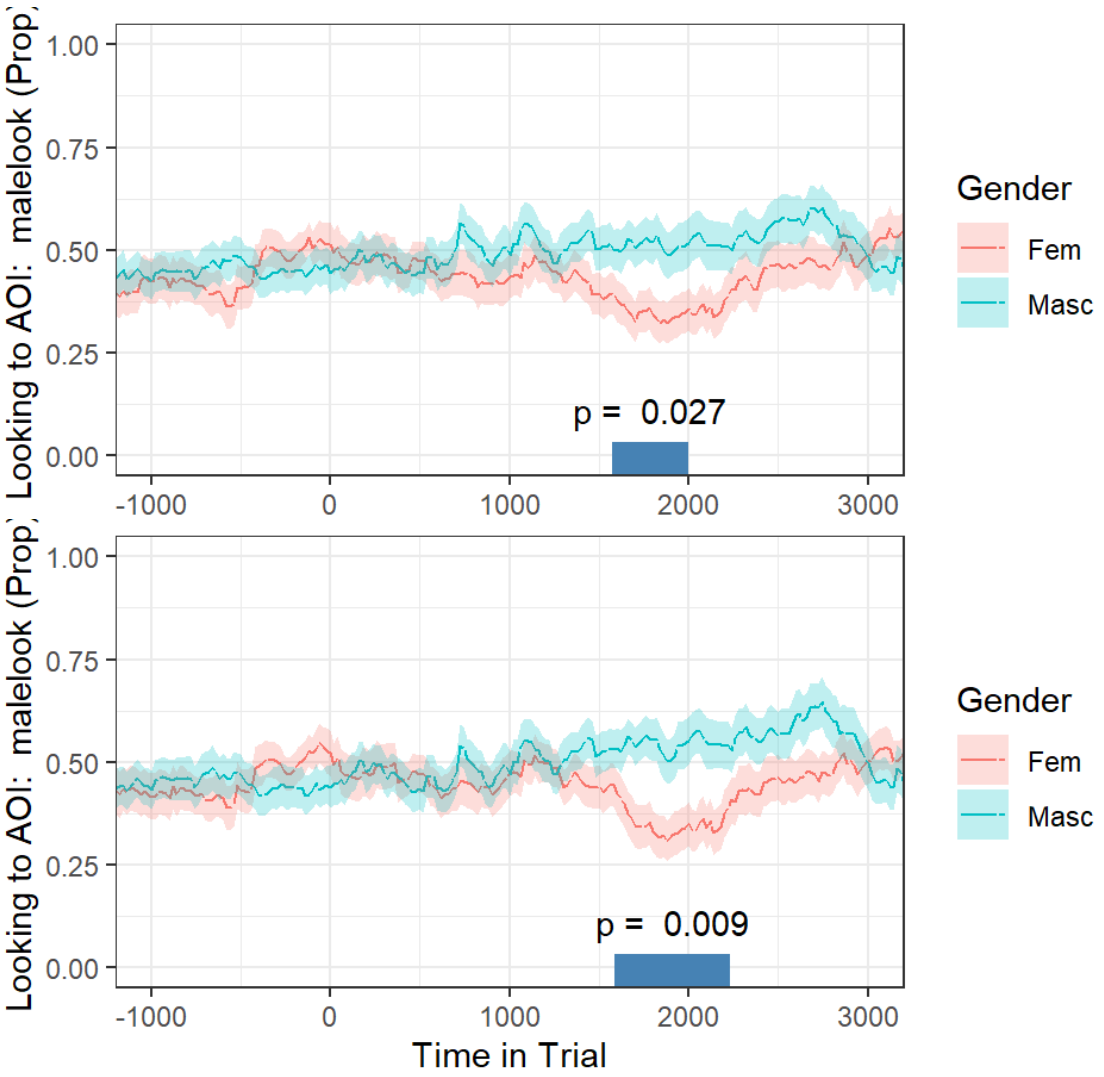
Results for Adjective 2

- Time windows
  - Significant interaction
- Permutation analyses
  - Significant cluster identified
  - p=0.027, with 37 kids with min. 2 trials/condition



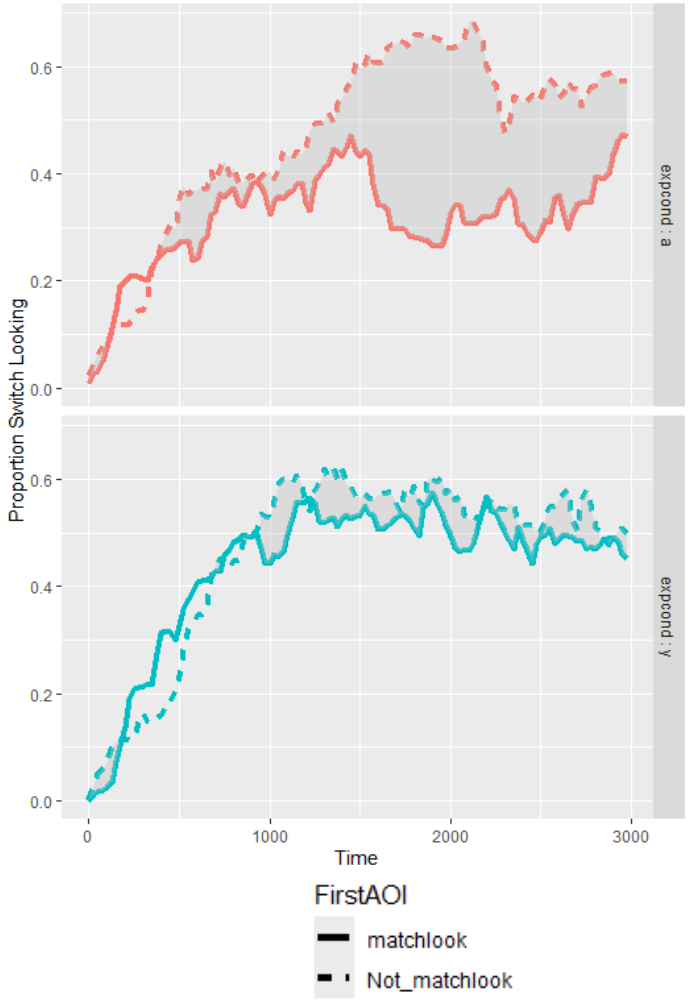
Results for Adjective 2

- Time windows
  - Significant interaction
- Permutation analyses
  - Significant cluster identified
  - p=0.027, with 37 kids with min. 2 trials/condition
  - p=0.009, with 31 kids with min. 3 trials/condition



Exploring the gender conditions

- The effect seems to be due to feminine stimuli only
  - Probably weaker response to masculine forms (condition y in the picture)
- Why? Is it generic masculine effect?
  - Unlikely at this age
- More familiarity with female faces
  - Would not explain why the grammar works for females and not for males
- Variability in standard vs. Prague/Bohemian-colloquial marking
  - velká – same in standard/colloquial
  - velkej – more common in colloquial Bohemian Czech; velký is neuter
- Could be tried with non-nom forms (hezkého, hezkou)



Discussion

- Why only the second adjective?
  - Children are slow
- It can take time to make the connection between grammar and social category

- Maybe we are tapping in a different kind of knowledge
  - than in the determiner/adjective – noun studies

Conclusions

- Czech 18- to 24-month-olds have emerging knowledge of adjective inflections
- They associate it with a semantic/social category
  - Contrasts with studies using informative determiners/adjectives before nouns
    - The current finding cannot be a lexically-specific association for a given noun – there was no noun
    - Gender agreement is likely to rely on a category representation
  - The inflection – semantics link
    - Both limitation and asset of the approach
    - Asset: shows that kids are ready to associate inflections with meaning differences, not just specific words
    - Limitation: is the gender category identified here the same as that used in agreement?
      - It is parsimonious to assume this, but should be tested

Thank you!

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